

Heterogeneous Markups Cyclicalities and Monetary Policy

Andrea Chiavari, Marta Morazzoni, and Danila Smirnov

Online Appendix

Contents

A Appendix	2
A.1 Data	2
A.2 Firm-Level Analysis: Robustness and Additional Results	3
A.2.1 Excluding the ZLB	3
A.2.2 Excluding Future Shocks	3
A.2.3 Linear Parametric Interaction	6
A.2.4 Alternative Monetary Policy Shocks	6
A.2.5 Using Founding Age by Jay Ritter	8
A.2.6 Grouping Firms by Sector and Quarter	9
A.2.7 Using Alternative Elasticities	9
A.2.8 Using Alternative Markups Measure	14
A.2.9 Excluding All Controls	17
A.2.10 Excluding All Fixed Effects	17
A.2.11 Using Dynamic Panel Data Approaches to Markups Measurement . .	17
A.2.12 Excluding the Financial Sector	20
A.2.13 Relative Response of Sales and COGS	20
A.3 Aggregate Implications: Additional Exercises	22
A.3.1 Direct and Indirect Effects	22
A.3.2 Sales-Weighted Markup Aggregation	27
A.3.3 Changing Firm-Level Responses Over Time	29

A Appendix

A.1 Data

This section provides details on the variables used in the analysis. First, following standard practices in the literature and ensuring that firms in our Compustat sample face the interest rate set by the Federal Reserve as their benchmark, we restrict our attention to firms that are incorporated in the U.S. Second, we exclude firms in utilities (SIC codes between 4900 and 4999) because they have heavily regulated prices.

Furthermore, we drop all the observations in our sample with missing industry classification, as well as those observations with negative or missing sales (**SALEQ**) and cost of goods sold (**COGSQ**). Whenever applicable, we deflate variables using a GDP deflator from the NIPA tables. Table A.1 reports summary statistics for the variables used in our regression analyses.

Table A.1: Summary Statistics

	Sales	Cogs	Assets	Leverage	Liquidity	Age (Compustat)	Age (Jay Ritter)	Markups
Mean	447.69	303.17	4919.69	0.45	0.16	9.46	23.61	1.78
P25	6.06	3.31	37.83	0.03	0.02	4	9	1.03
P50	31.00	17.18	229.50	0.18	0.07	8	16	1.30
P75	164.58	100.60	1118.33	0.39	0.22	14	29	1.86
N	715,874	715,874	685,784	641,316	683,696	715,874	146,112	715,874

Note: Summary statistics of cleaned quarterly Compustat between 1990q1 and 2016q4. Sales, Cogs, and Assets are measured in millions of real 2012 U.S. dollars, while Leverage and Liquidity are ratios and Age is in years since IPO (Compustat) and since foundation (Jay Ritter).

We exploit information on sales (**SALEQ**) to measure firm-level production, while we use the cost of goods sold (**COGSQ**) to determine the variable inputs used in production, and gross capital (**PPEGTQ**) to measure tangible capital. In line with the literature, we use selling, general, and administrative expenses (**XSGAQ**) as a measure of overhead costs, and the variable **ATQ** as a measure of total assets. Finally, we use cash and short-term investments (**CHEQ**) and short and long-term liabilities (**DLCQ** and **DLTTQ**) to compute liquidity and leverage

(i.e. CHEQ/ATQ and DLCQ/ATQ + DLTTQ/ATQ respectively). As explained in Section 3, our benchmark measure for markups is the ratio between sales and cost of goods sold, multiplied by the input-output elasticities computed by [De Loecker, Eeckhout and Unger \(2020\)](#).

A.2 Firm-Level Analysis: Robustness and Additional Results

A.2.1 Excluding the ZLB

In what follows, we estimate Equation (5) between 1990q1 and 2008q4, thereby excluding the Zero Lower Bound (ZLB) period. We do this for several reasons: during the ZLB period, (i) the measure of monetary policy (MP) shocks from [Jarociński and Karadi \(2020\)](#) exhibits nearly no variation, and (ii) central banks favored new unconventional monetary policy practices over standard ones. Both reasons suggest that the ZLB period could be atypical and we want to ensure that this is not driving our main insights. Figure [A.1](#) shows the result. Overall, we notice that excluding the ZLB from the period of analysis does not affect our qualitative conclusions, but increases the magnitude and significance of the estimated coefficient $\widehat{\gamma_{age,h}^0}$.

A.2.2 Excluding Future Shocks

In the following robustness analysis, we estimate an alternative specification of Equation (5) in which we do not include future monetary policy shocks as controls, and that is given by:

$$(A.1) \quad \begin{aligned} \Delta_h \log \mu_{i,t+h} = & \sum_{x \in \mathcal{X}} \left(\alpha_{x,h} + \beta_{x,h} \Delta Y_{t-1} + \sum_{k=0}^{\kappa} \gamma_{x,h}^k \varepsilon_{t-k}^m \right) \times \mathbf{1}_{i \in \mathcal{I}^x} \\ & + \sum_{\ell=1}^L \boldsymbol{\delta}'_h \mathbf{X}_{i,t-\ell} + \varphi_{i,h} + \varphi_{s,t,h} + \boldsymbol{\vartheta}_h \mathbf{t} + u_{i,t+h}, \end{aligned}$$

with horizons $h = 0, 1, \dots, H$. Figure [A.2](#) shows the result. Overall, excluding future MP shocks from Equation (5) does not qualitatively affect our results, but it increases the downward bias present in LP models, consistent with the insights in [Teulings and Zubanov](#)

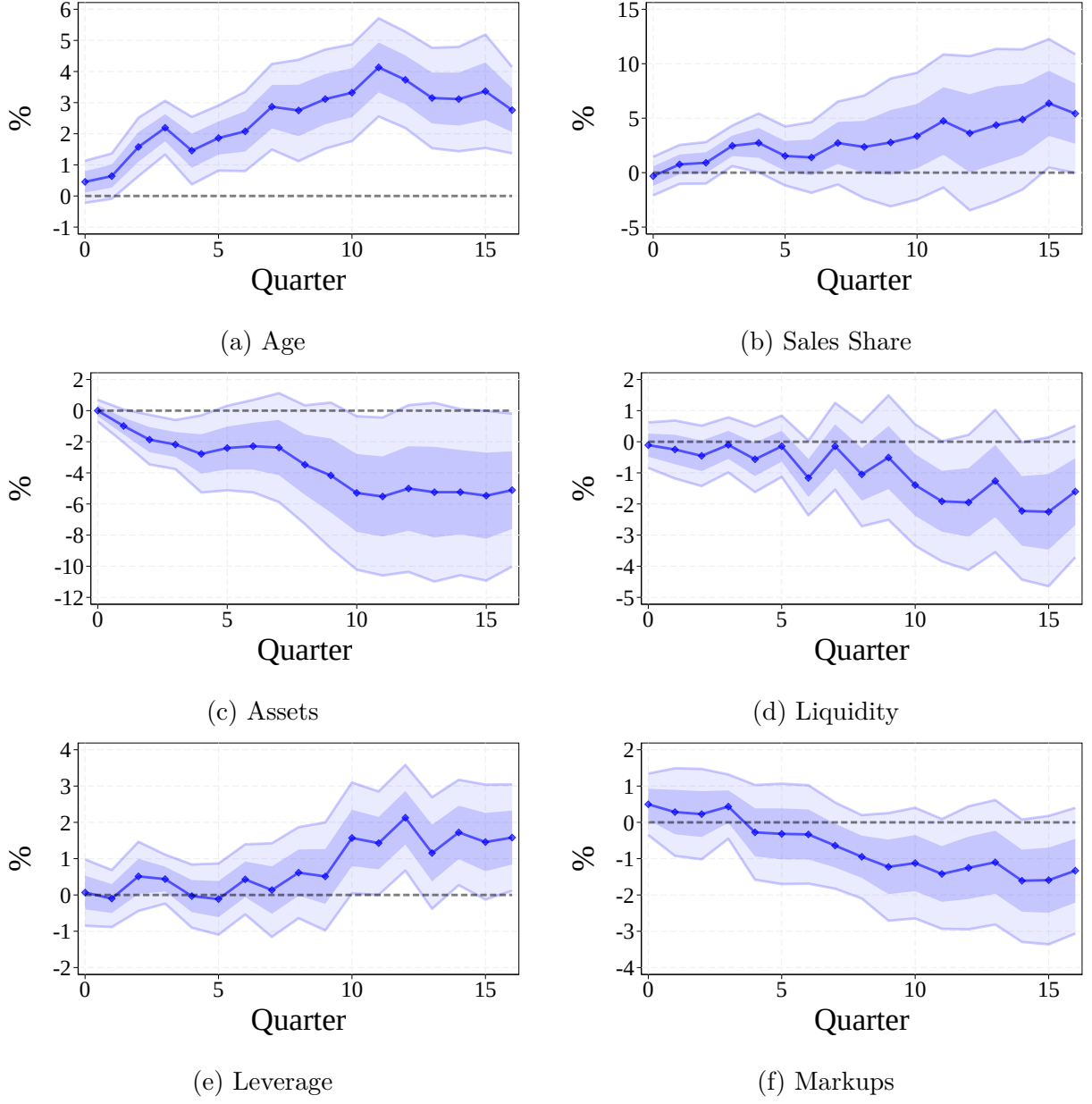


Figure A.1: Heterogeneous Markups Cyclicity - No Zero Lower Bound Period
Note: Figure A.1 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, limiting the sample between 1990q1 and 2008 q4 (before the zero lower bound period). Figure A.1 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

(2014).

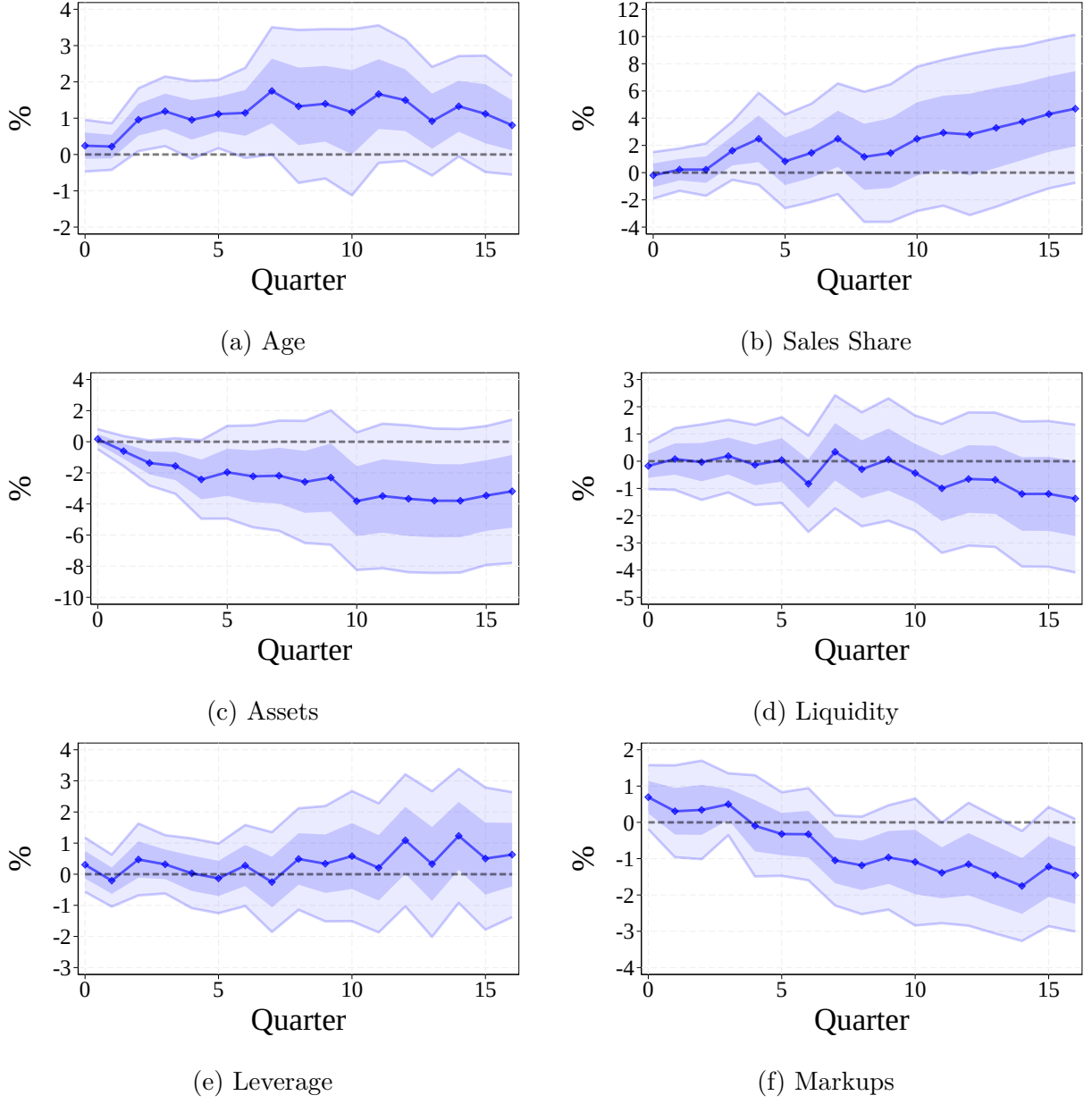


Figure A.2: Heterogeneous Markups Cyclicity - Excluding Future Shocks

Note: Figure A.2 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates except for future MP shocks (note that the baseline estimation contains instead 4 leads), and for horizons $h = 1, \dots, 16$. Figure A.2 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

A.2.3 Linear Parametric Interaction

In the following exercise, we estimate an alternative specification of Equation (5) in which we use a linear parametric approach, similarly in spirit to the analysis in [Ottonello and Winberry \(2020\)](#). Instead of characterizing firms with dummies, we linearly interact the MP shock series with measures of firm age, assets, sales share, leverage, liquidity and markup, and estimate:

$$(A.2) \quad \begin{aligned} \Delta_h \log \mu_{i,t+h} = & \sum_{x \in \mathcal{X}} \left(\alpha_{x,h} + \beta_{x,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{x,h}^k \varepsilon_{t+k}^m \right) \times x_{t-1} \\ & + \sum_{\ell=1}^L \boldsymbol{\delta}'_h \mathbf{X}_{i,t-\ell} + \varphi_{i,h} + \varphi_{s,t,h} + \boldsymbol{\vartheta}_h \mathbf{t} + u_{i,t+h}, \end{aligned}$$

for horizons $h = 0, 1, \dots, H$. Figure [A.3](#) shows our results. Overall, we notice that the coefficients estimated through Equation (A.2) still reflect the main insights of our benchmark specification. Specifically, Figure [A.3](#) confirms that older firms show more countercyclical markups in response to MP shocks, while there is no evident heterogeneity in the response of markups by firms' size, sales share, leverage, liquidity, or the level of their markup.

A.2.4 Alternative Monetary Policy Shocks

Here, we estimate Equation (5) using the alternative monetary policy shock measure from [Gürkaynak, Sack and Swanson \(2005\)](#). Their series is similar to the one in [Jarociński and Karadi \(2020\)](#), but it does not account for the information channel of monetary policy.

Figure [A.4](#) shows the results. Overall, using alternative MP shocks by [Gürkaynak, Sack and Swanson \(2005\)](#) produces a set of impulse response functions (IRFs) that is both qualitatively and quantitatively close to the benchmark one, with stronger statistical significance.

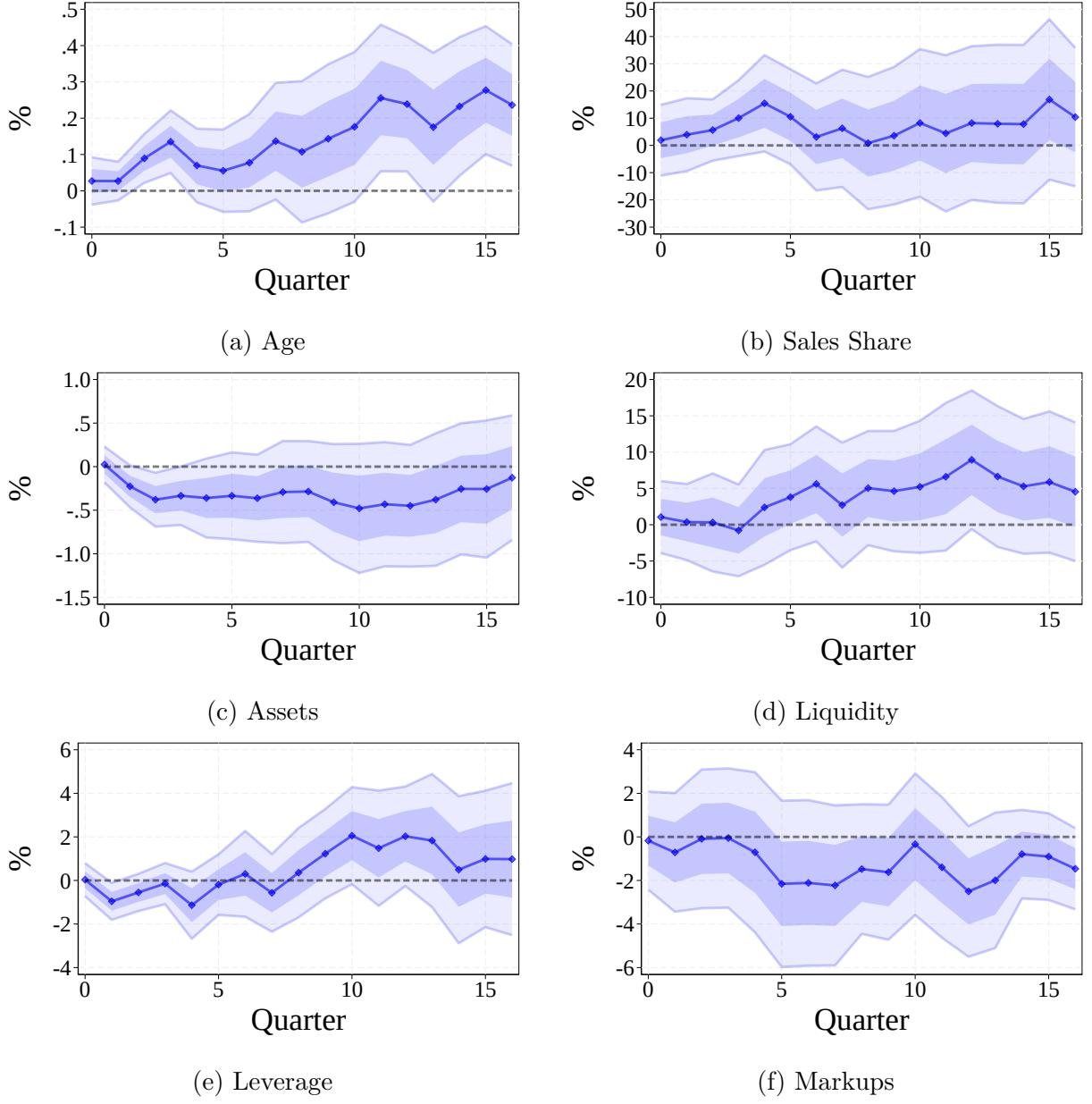


Figure A.3: Heterogeneous Markups Cyclicity - Linear Parametric Specification

Note: Figure A.3 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (A.2) using a linear parametric approach, with all covariates and for horizons $h = 1, \dots, 16$. Figure A.3 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

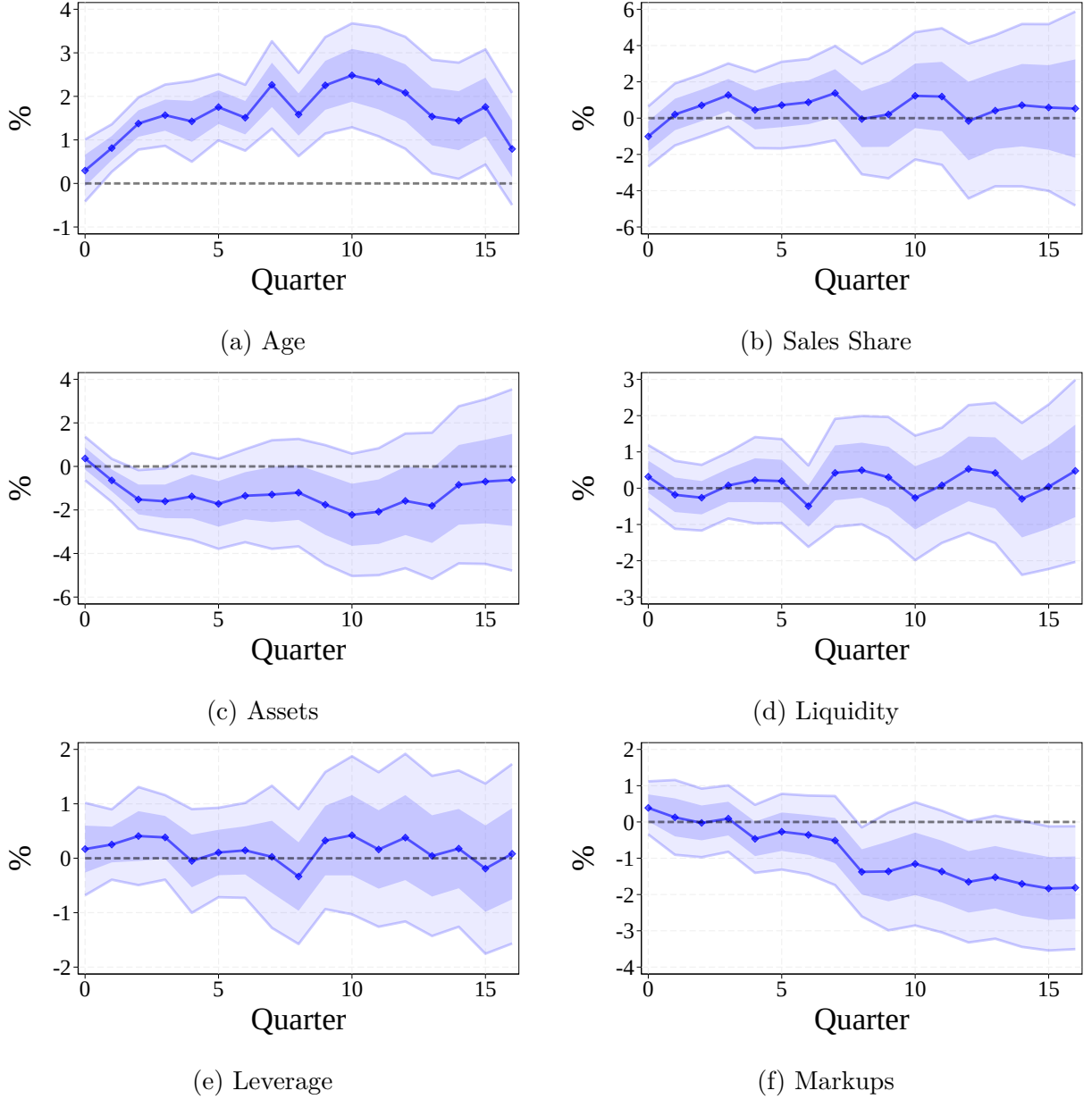


Figure A.4: Heterogeneous Markups Cyclicity - Gürkaynak, Sack & Swanson (2005)
Note: Figure A.4 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. We use the MP shocks series exogenously identified in Gürkaynak, Sack and Swanson (2005). Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.4 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

A.2.5 Using Founding Age by Jay Ritter

In the robustness analysis that follows, we estimate Equation (5) using the "true" founding age of firms instead of corporate age, which is made available by Jay Ritter for a subsample

of Compustat (about 20 percent of the observations approximately, hence a significant empirical limitation if we were to use only founding age in our baseline estimation). Results are shown in Figure A.5. Overall, we see that using this alternative measure of firm age does not significantly alter our main conclusions. In particular, older firms keep their markups relatively higher compared to young ones after the arrival of an MP shock. Neither leverage, liquidity or absolute size (measured in assets) are strong predictors of heterogeneity in markup responses to a monetary tightening, as is also the case for sales share and the markup level.

A.2.6 Grouping Firms by Sector and Quarter

In the following exercise, we estimate Equation (5) using a different definition for firms' groups $\mathbf{1}_{i \in \mathcal{I}^x}$. In the main analysis, we define firms' dummies by being above or below the median of $x \in \mathcal{X} = \{age, sales\ share, leverage, liquidity, assets\}$, considering the entire sample and according to firms' characteristics in the previous year. Here, we instead define firms' dummies by being above or below the median of those same variables but within a given sector and a given quarter of the previous year. Figure A.6 shows the result. Overall, using a different definition of $\mathbf{1}_{i \in \mathcal{I}^x}$ does not change our conclusions, suggesting that our results do not depend on the particular definition of the dummies capturing firm-level heterogeneity.

A.2.7 Using Alternative Elasticities

In the following robustness exercise, we estimate Equation (5) using firm-level markups calculated with different production function elasticities. In particular, instead of using the elasticities provided by De Loecker, Eeckhout and Unger (2020), which are common for all the quarters within a year, we calculate our own input-output elasticities allowing them to vary at a quarter level. In particular, we estimate (i) a Cobb-Douglas production function which varies at quarter and sector levels, and (ii) a Translog production function which also varies at quarter and sector levels. Note that this second specification also allows us to

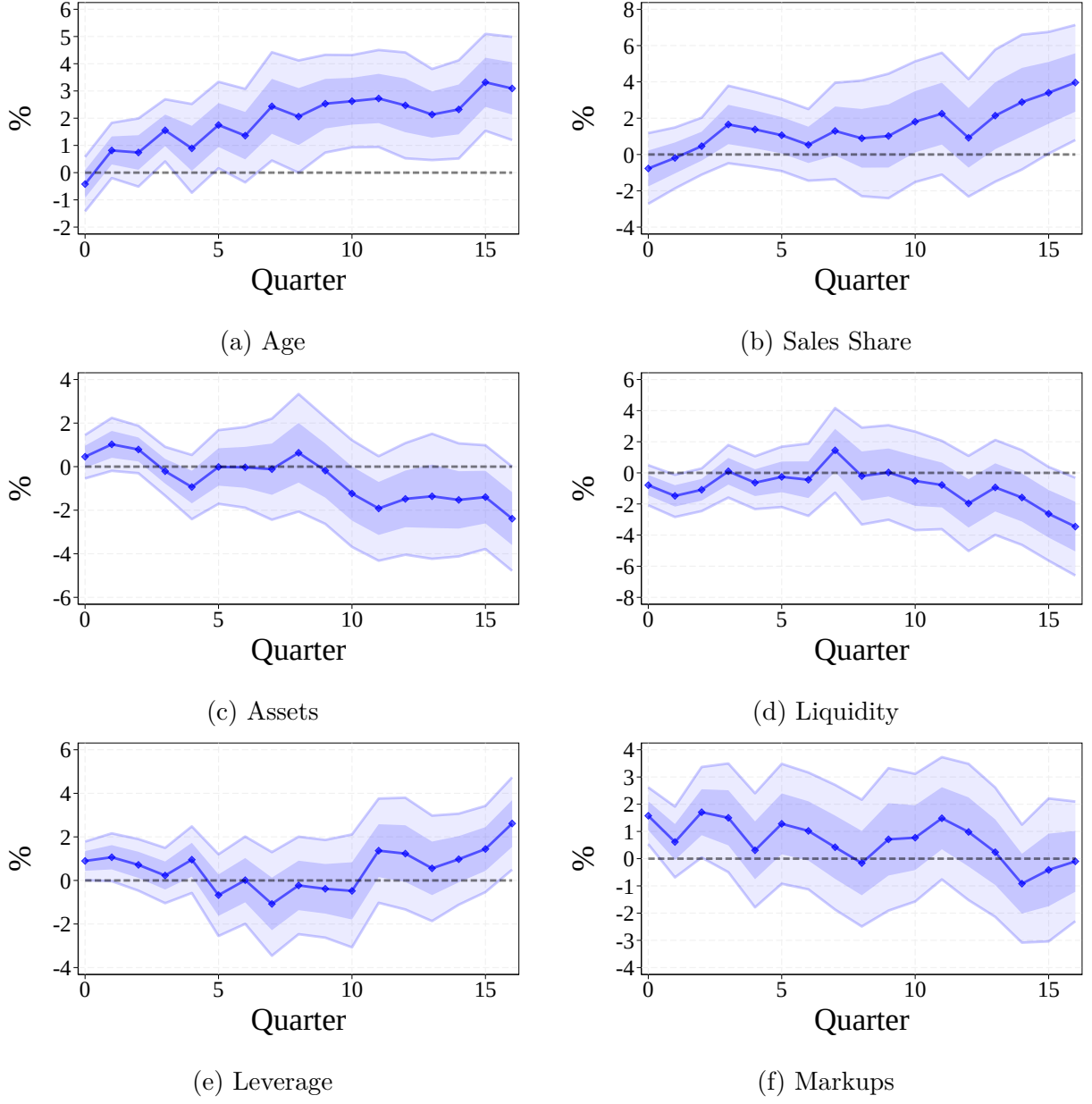


Figure A.5: Heterogeneous Markups Cyclicity - Using Founding Age

Note: Figure A.5 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. We use the founding age of firms in Compustat as collected in the database by Jay Ritter. Coefficients $\widehat{\gamma_{x,h}^0}$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.5 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma_{x,h}^0}$. Standard errors clustered at the firm and quarter level.

have production function elasticities that are heterogeneous across firms within sectors and

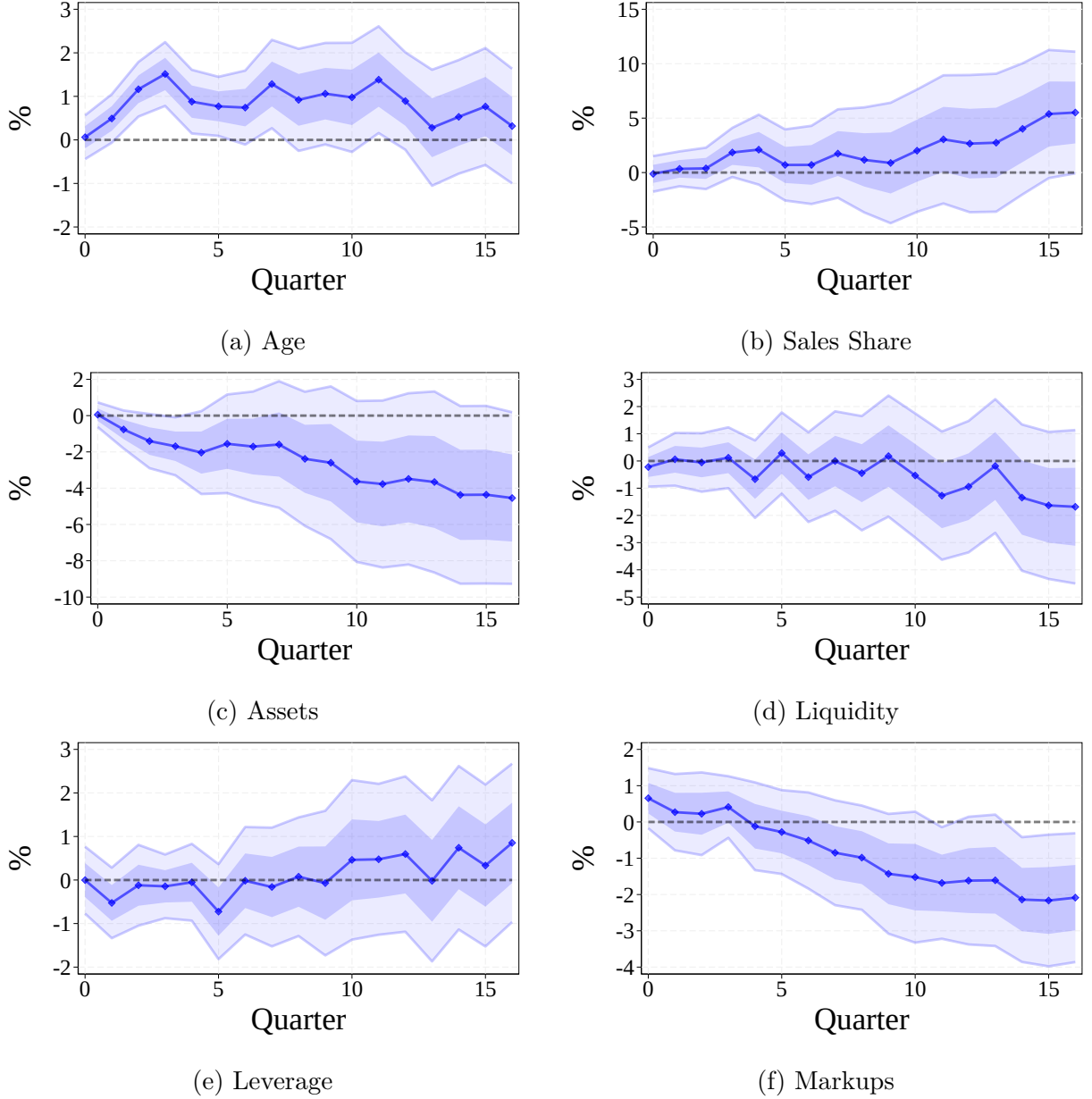


Figure A.6: Heterogeneous Markups Cyclicity - Sector/Quarter Dummies

Note: Figure A.6 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. We construct dummies equal to one if the variable is above the median within a sector and quarter. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.6 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

quarters.¹

¹The elasticity coming from the Translog production function is a function of firm-level capital stock (PPEGT in Compustat). For data quality, we interpolate between adjacent points our estimates whenever this

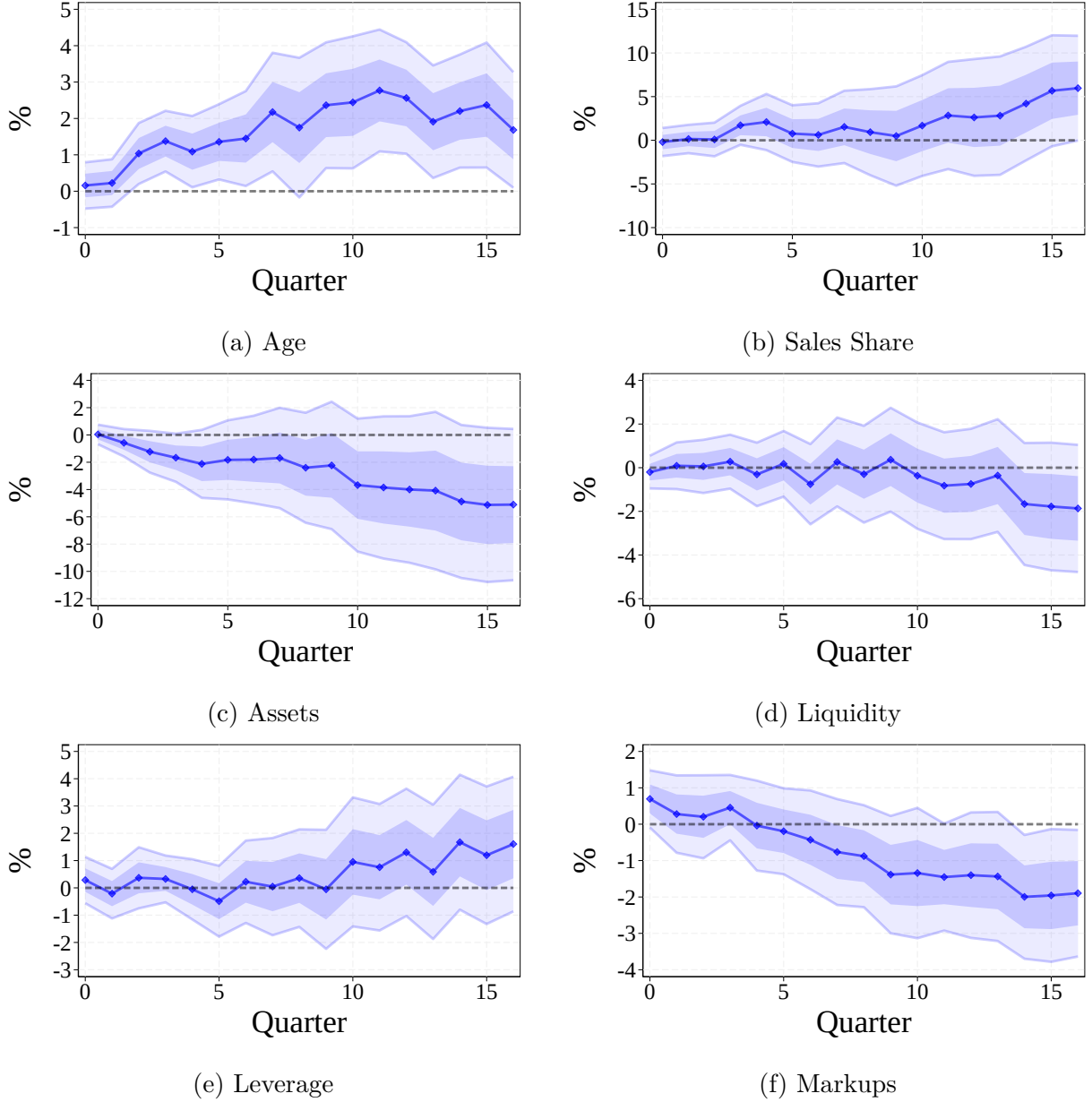


Figure A.7: Heterogeneous Markups Cyclicity - Cobb-Douglas Elasticity

Note: Figure A.7 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. When measuring firm-level markups, we calculate the input-output elasticity $\theta_{s,t}^v$ in Equation (4) by estimating a Cobb-Douglas production function that varies at quarter and sector-level. Coefficients $\widehat{\gamma_{x,h}^0}$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.7 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma_{x,h}^0}$. Standard errors clustered at the firm and quarter level.

Results are shown in Figure A.7 and A.8. The Cobb-Douglas specification shows similar item is missing. We check that this does not affect results, although it improves estimation precision.

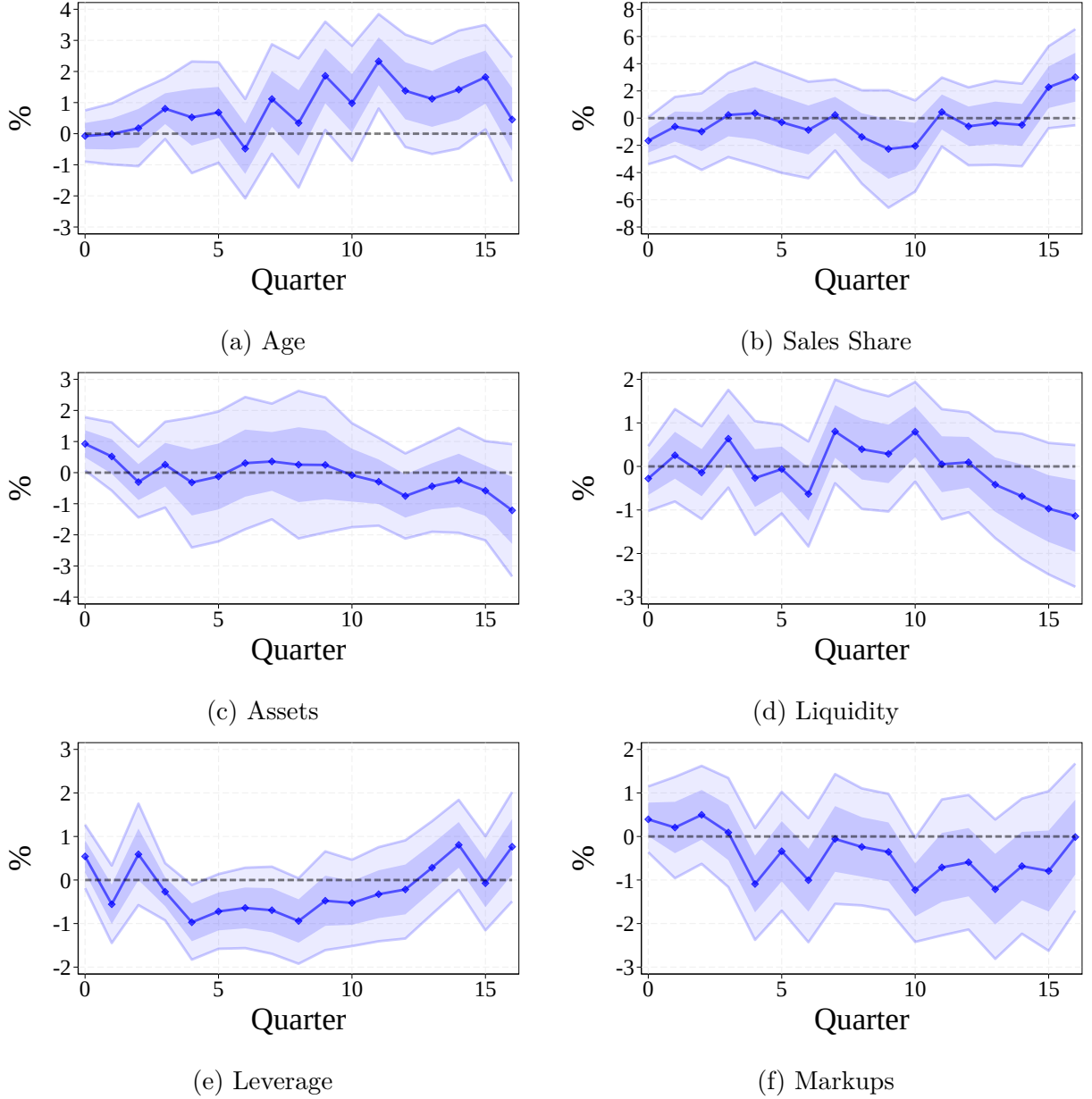


Figure A.8: Heterogeneous Markups Cyclicalty - Translog Elasticity

Note: Figure A.8 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. When measuring firm-level markups, we calculate the input-output elasticity $\theta_{s,t}^v$ in Equation (4) by estimating a Translog production function that varies at quarter and sector level. Coefficients $\widehat{\gamma_{x,h}^0}$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.8 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma_{x,h}^0}$. Standard errors clustered at the firm and quarter level.

patterns regarding the magnitude and significance of the estimated coefficients compared to our benchmark. The Translog specification shows instead much noisier estimates, and

the IRF of relatively older firms is significant at the 90 percent level only in quarter 11. Admittedly, estimating Translog production functions imposes heavy requirements on the data.

The somewhat mixed results of this specific robustness check call for a more cautious interpretation of the main findings regarding the different cyclicalities of markups by firm-specific characteristics. However, they still suggest that the firm-level heterogeneous markup cyclicalities we uncovered may not be just driven by heterogeneity in firm-level input-output elasticities, but also by heterogeneity in the firm-level ratio of sales to variable costs.

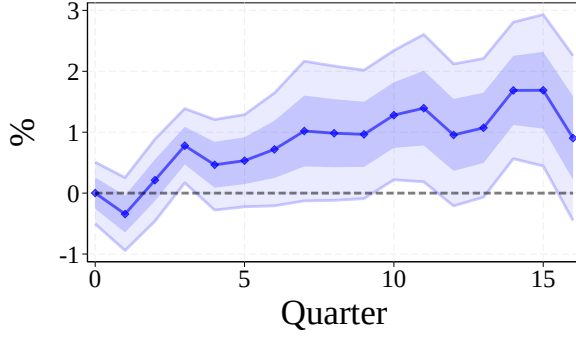
A.2.8 Using Alternative Markups Measure

In this exercise, we estimate again Equation (5) using firm-level markups calculated with a different methodology compared to that presented in Equation (4). In particular, we follow the "accounting-profit approach" adopted by [Baqaee and Farhi \(2020\)](#), where markups are:

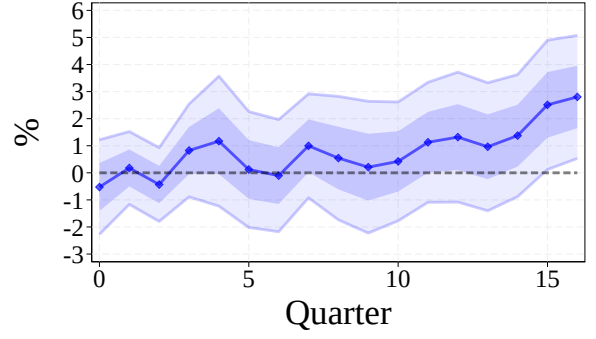
$$(A.3) \quad \begin{aligned} \mu &= \frac{v}{1 - \ell}, \\ \ell &= \frac{\text{OIBDPQ} - \text{DPQ}}{\text{SALEQ}}, \end{aligned}$$

v are the returns to scale in the firm's technology, and $\text{OIBDPQ} - \text{DPQ}$ denotes operating income before depreciation, net of depreciation. We first compute this measure — also known as the Lerner Index — imposing constant returns to scale across all firms, namely $v = 1$. Second, we allow for returns to scale to be heterogeneous across firms, and informed by the Translog production-function estimation we have carried out in Section [A.2.7](#).

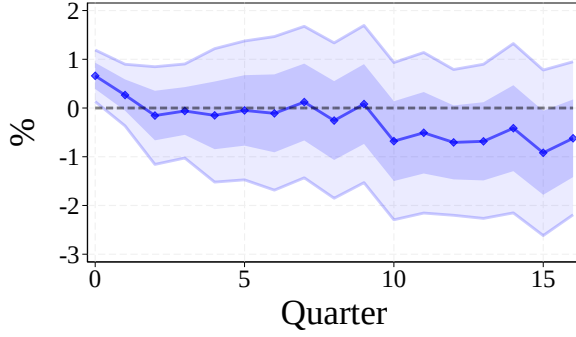
Results are shown in Figure [A.9](#) and [A.10](#). Overall, using these alternative measures of market power, often considered second-best compared to our main measure, does not affect the qualitative implications of our results. We stress though that it dampens the magnitude and statistical significance of the firm-age coefficients compared to our benchmark estimates from the main analysis (i.e. coefficients are significant at the 90 percent level only in few quarters).



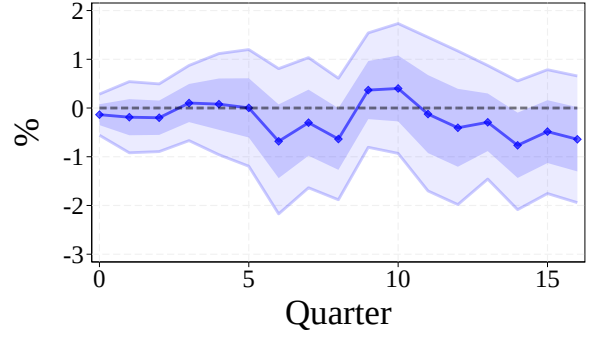
(a) Age



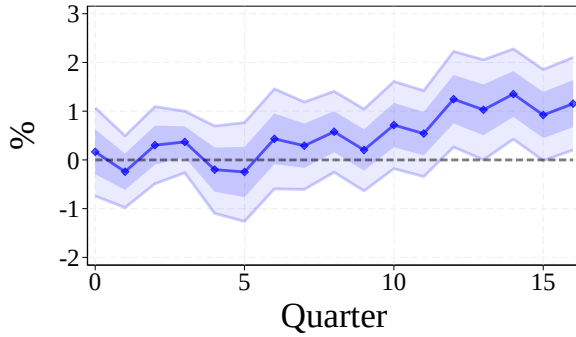
(b) Sales Share



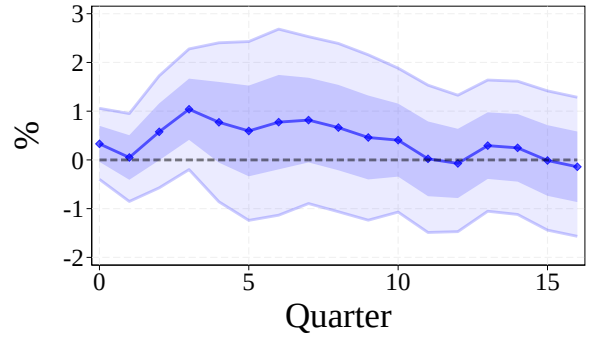
(c) Assets



(d) Liquidity



(e) Leverage



(f) Markups

Figure A.9: Heterogeneous Markups Cyclicalities - Alternative Markup Measure

Note: Figure A.9 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. We use the Lerner index as an alternative measure to firms' markups, as defined in Equation (A.3). Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.9 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

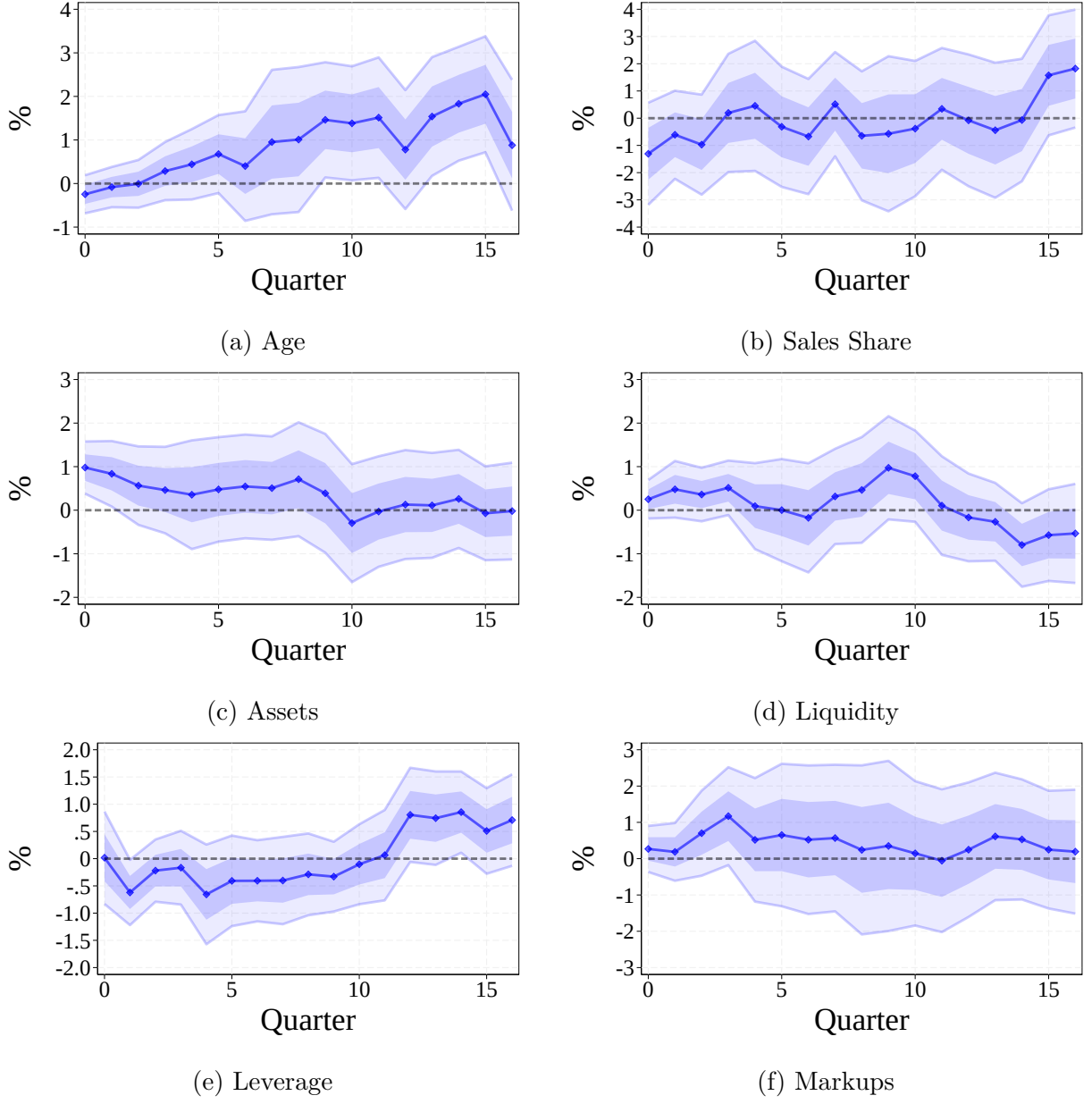


Figure A.10: Heterogeneous Markups Cyclicity - Alternative Markup Measure without Constant Returns to Scale

Note: Figure A.10 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. We use the Lerner index as an alternative measure to firms' markups, as defined in Equation (A.3) and exploiting the Translog returns to scale estimated in A.8. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure A.9 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

A.2.9 Excluding All Controls

In this exercise, we estimate again Equation (5) but excluding most of the controls from the analysis, to show that results are not driven by their inclusion. Specifically, we remove the interaction of the MP shock series with past GDP, as well as aggregate and firm-level controls. Recall that macro-level controls include GDP and CPI growth, changes in the 1-year Treasury rate, and the EBP (and 4 lags for each variable). Firm-level controls comprise sales growth and overhead costs relative to sales (and 4 lags for each variable). Results are presented in [Figure A.11](#) and show no major qualitative change to our preferred specification.

A.2.10 Excluding All Fixed Effects

In this exercise, we estimate again Equation (5) but excluding both firm and sector-time fixed effects, to show that results are not driven by their inclusion. Results are presented in [Figure A.12](#) and show no significant qualitative change to our preferred specification.

A.2.11 Using Dynamic Panel Data Approaches to Markups Measurement

Although in the main text we used output elasticities estimated with structural production function estimators following the approach in [De Loecker et al. \(2020\)](#), these estimators face their own challenges in the context of market power. These challenges arise both in the first-stage inversion (e.g., [Bond et al., 2021](#)) and when estimating gross output production functions (e.g., [Flynn et al., 2019](#), [Gandhi et al., 2020](#)).

Because of this, here we exploit dynamic panel data approaches that do not require the first stage inversion, at the cost of assuming log-linear productivity dynamics. In particular, we estimate the output elasticities of the production function using [Arellano and Bond \(1991\)](#) with non-constant and constant returns to scale and [Blundell and Bond \(1998\)](#) with non-constant returns to scale.

To address this, we exploit dynamic panel data approaches that do not require the first-stage inversion, at the cost of assuming log-linear productivity dynamics. Specifically, we

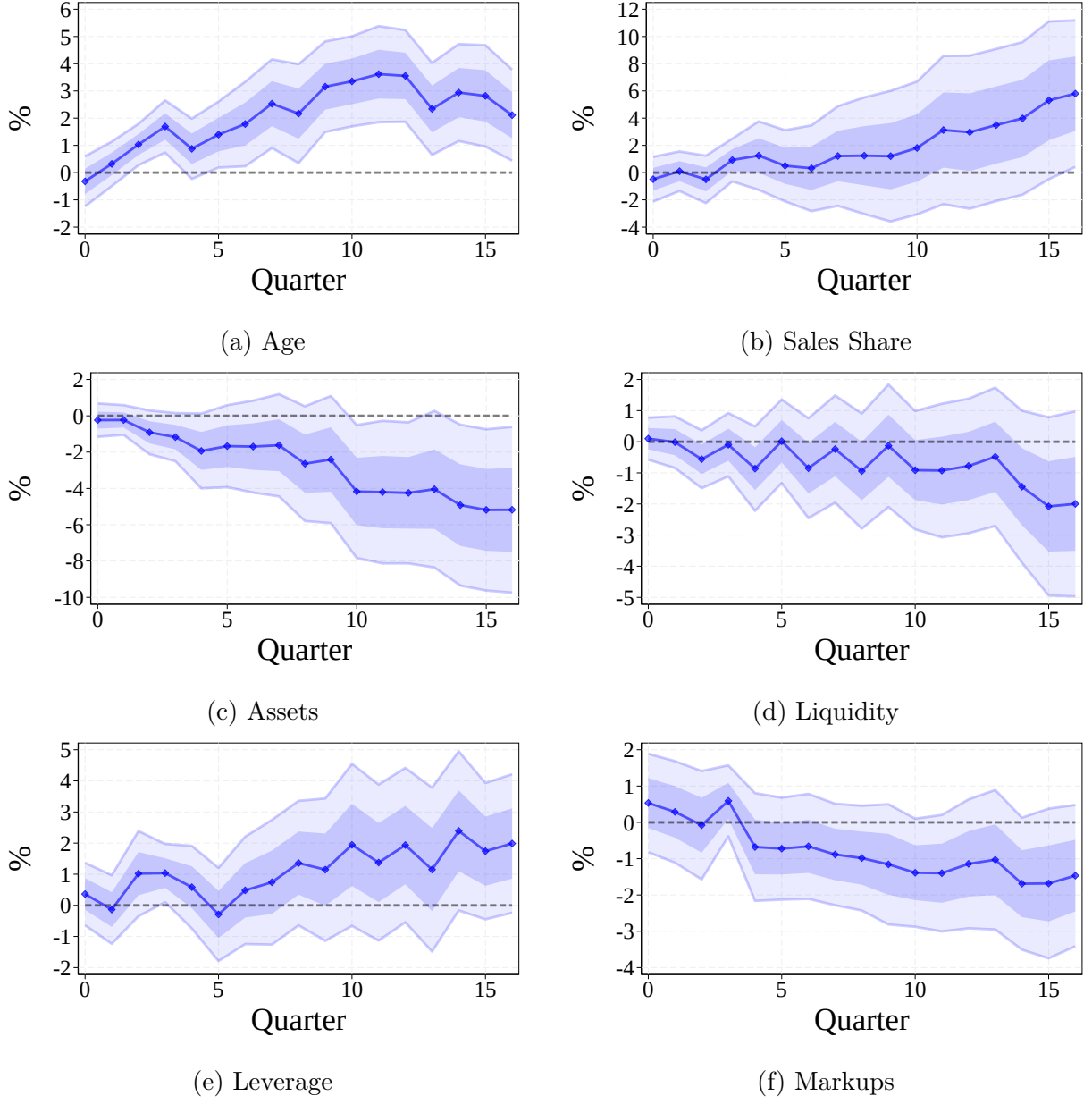


Figure A.11: Heterogeneous Markups Cyclicity - No Controls

Note: Figure A.11 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma_{x,h}^0}$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, but without any control. Figure A.11 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma_{x,h}^0}$. Standard errors clustered at the firm and quarter level.

estimate the output elasticities of the production function using Arellano and Bond (1991) with both non-constant and constant returns to scale, and Blundell and Bond (1998) with

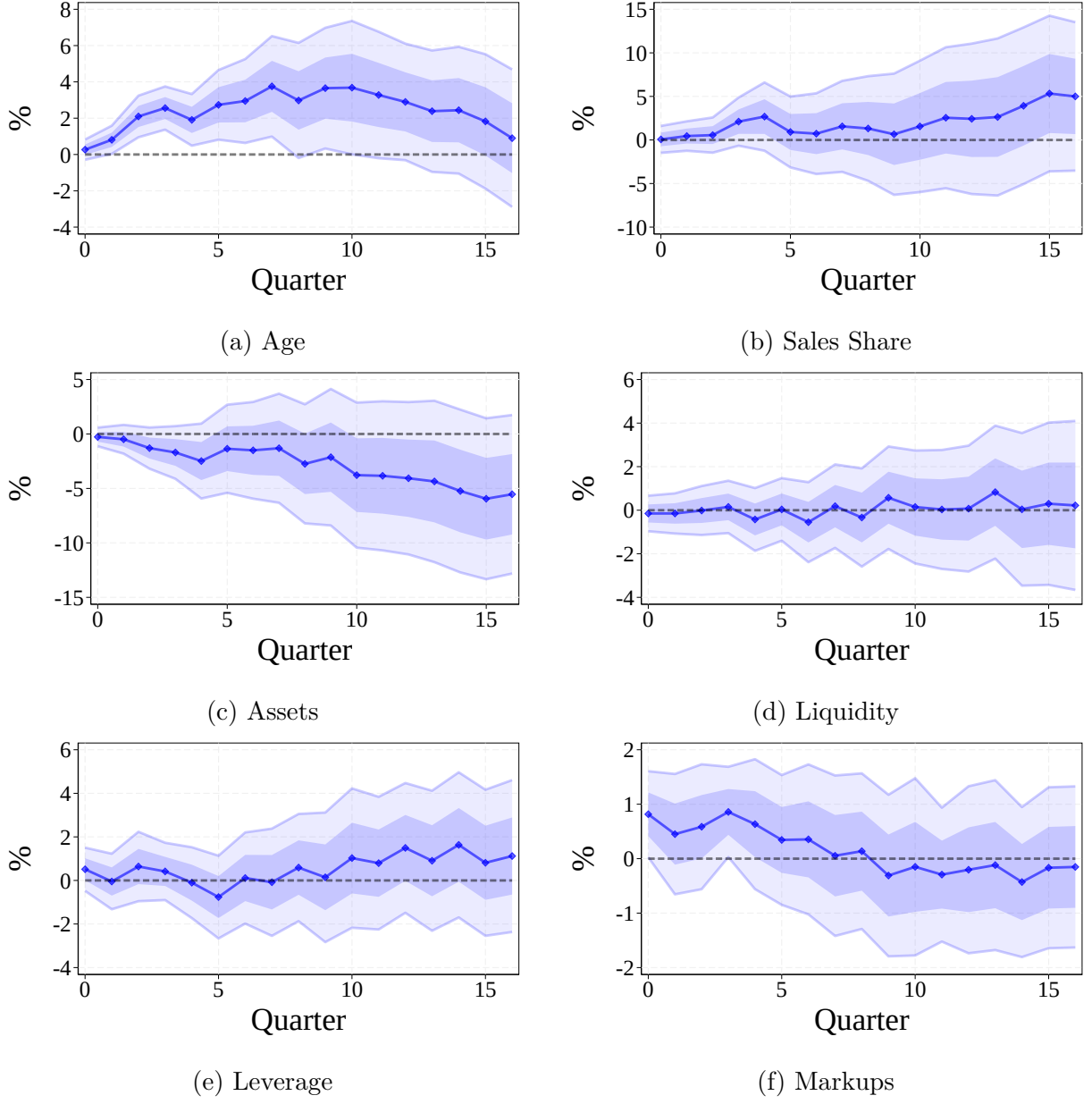


Figure A.12: Heterogeneous Markups Cyclicity - No Fixed Effects

Note: Figure A.12 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, but without any FE. Figure A.12 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

non-constant returns to scale.

Figure A.13 shows the results for the output elasticity estimated using Arellano and Bond

(1991) with non-constant returns to scale. Figure A.14 shows the results for the output elasticity estimated using Arellano and Bond (1991) with constant returns to scale. Figure A.15 shows the results for the output elasticity estimated using Blundell and Bond (1998) with non-constant returns to scale. Overall, we find that all these different specifications yield results quantitatively similar to our baseline specification in the main text.

A.2.12 Excluding the Financial Sector

In this robustness exercise, we re-estimate Equation (5) on the full sample after dropping financial firms (those with SIC codes between 6000 and 6999). Financial firms differ from the rest of the economy in how their markups are measured and in how they respond to monetary policy, so we verify that they are not driving our results. Figure A.16 shows the result. Excluding financial firms leaves our qualitative conclusions unchanged.

A.2.13 Relative Response of Sales and COGS

Figure A.17 and Figure A.18 present additional results on the relative response of sales and cost of goods sold — the two main components of our firm-level markup estimator — across various firm characteristics, conditional on a MP shock. Despite its weak statistical significance (IRFs are significant at the 90 percent level only in few quarters), we find that firm age is the most relevant predictor of heterogeneity in response to MP shocks for both sales and cost of goods sold. Moreover, while both variables show a negative response, the cost of goods sold declines more strongly and significantly for old firms, compared to firm sales, which is consistent with the relative increase of markups for old firms conditional on a contractionary MP shock.

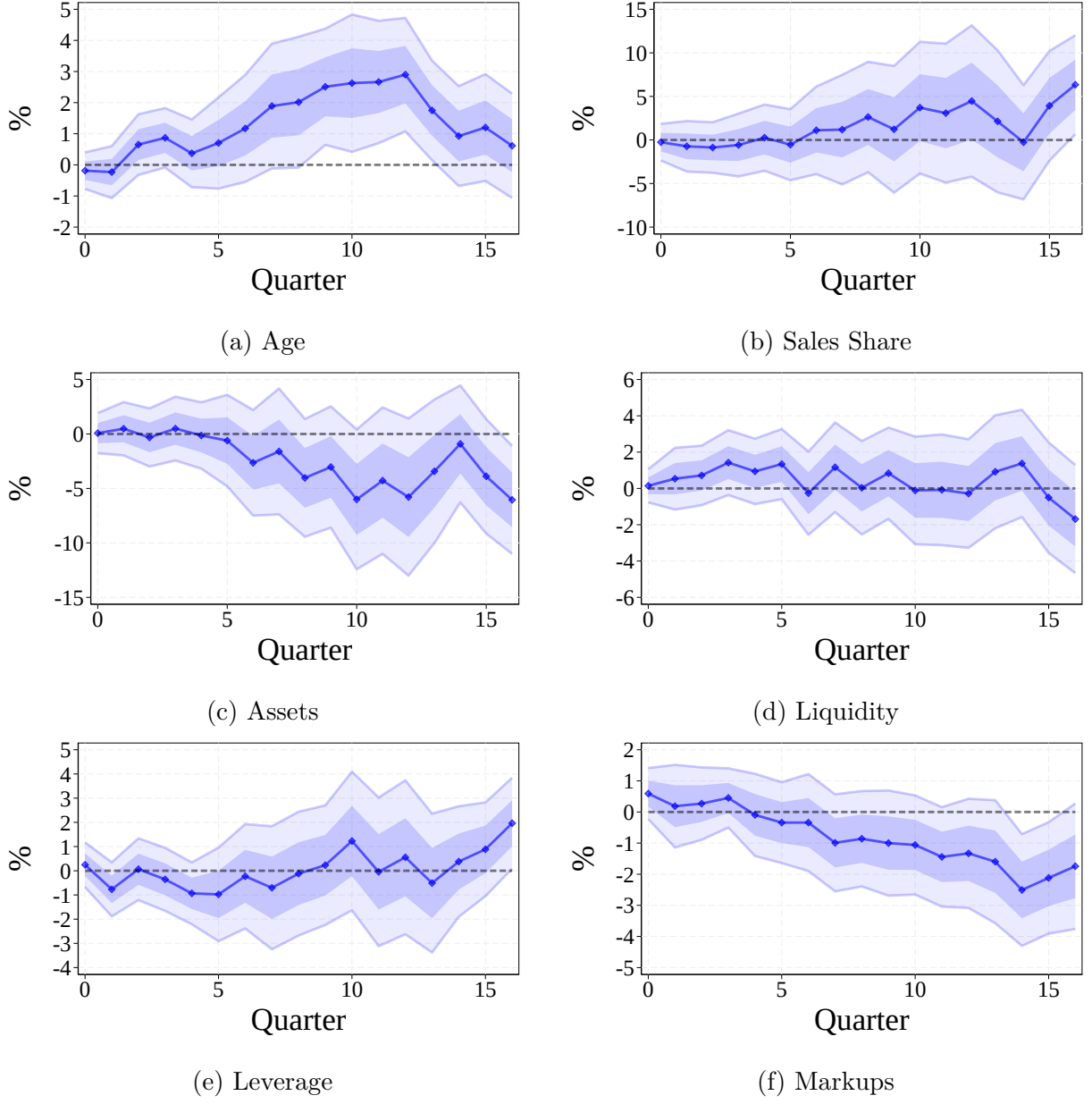


Figure A.13: Heterogeneous Markups Cyclicity - Arellano Bond Estimator w/o CRTS
Note: Figure A.13 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, limiting the sample between 1990q1 and 2008 q4 (before the zero lower bound period). Figure A.13 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

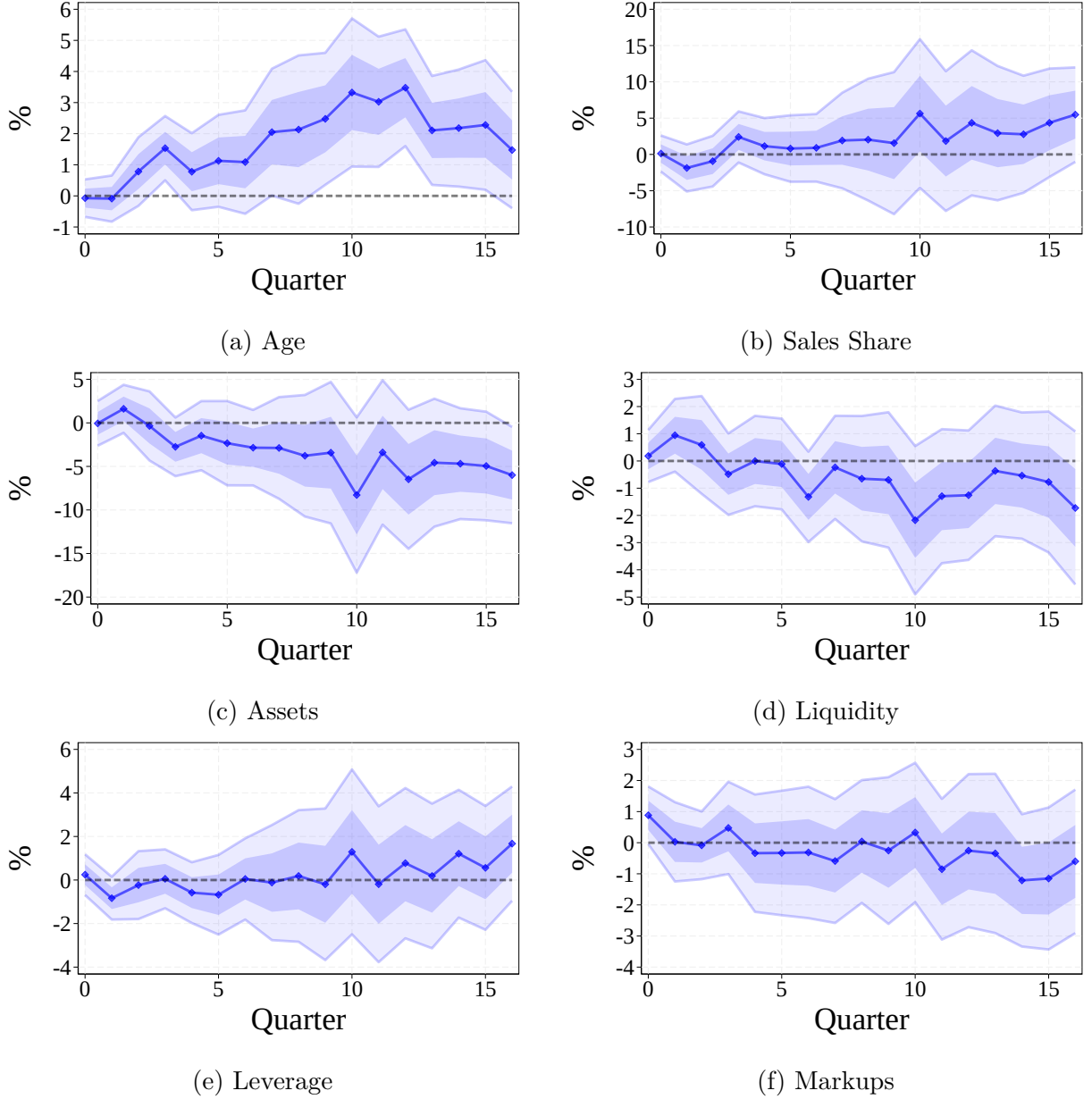


Figure A.14: Heterogeneous Markups Cyclicity - Arellano Bond Estimator w/ CRTS
Note: Figure A.14 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, limiting the sample between 1990q1 and 2008 q4 (before the zero lower bound period). Figure A.14 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

A.3 Aggregate Implications: Additional Exercises

A.3.1 Direct and Indirect Effects

In what follows, we decompose the IRF estimated in Figure 4 to highlight the contribution of both the direct and indirect effect of firm-level dynamics to the conditional cyclicity of

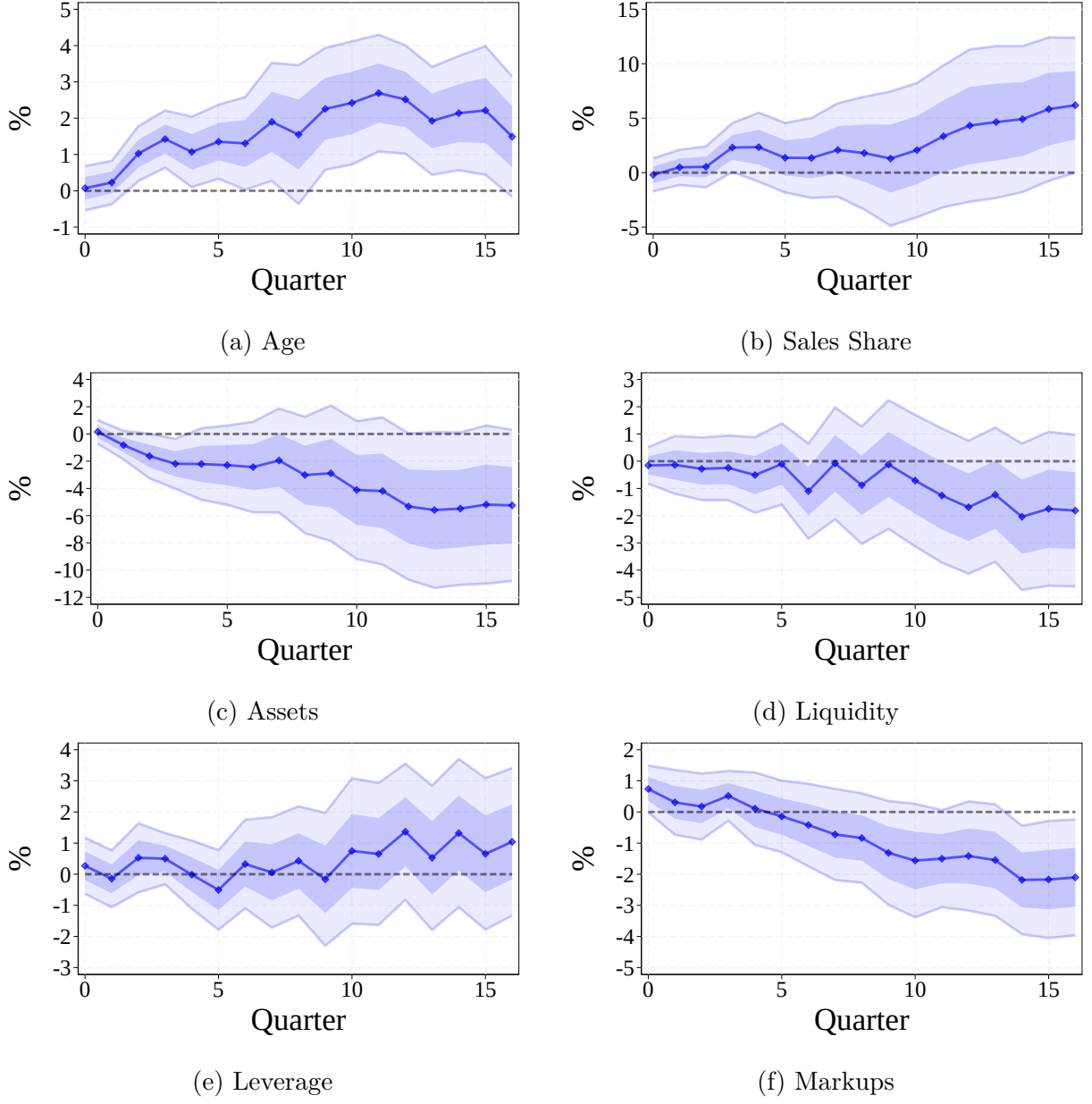


Figure A.15: Heterogeneous Markups Cyclicity - Blundell Bond Estimator w/o CRTS
Note: Figure A.15 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, limiting the sample between 1990q1 and 2008 q4 (before the zero lower bound period). Figure A.15 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

the aggregate markup, as specified in Equation (9). Interestingly, Figure A.19 clarifies that the heterogeneous impact on markups across different firm-age groups is what quantitatively

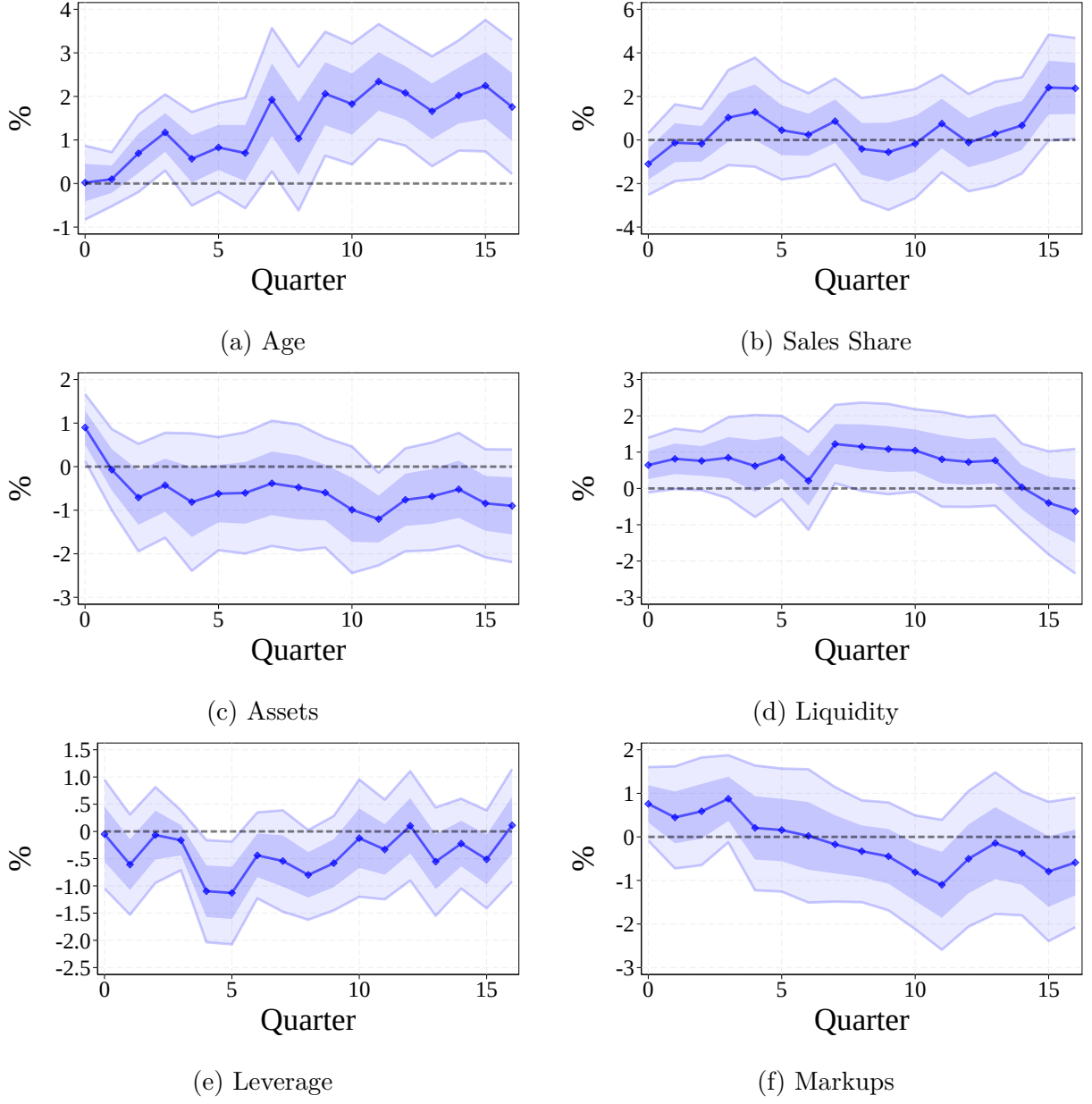


Figure A.16: Heterogeneous Markups Cyclicity - Excluding the Financial Sector
Note: Figure A.16 shows the relative markup response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with all covariates and for horizons $h = 1, \dots, 16$, dropping financial firms (SIC codes 6000–6999). Figure A.16 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

matters for the aggregate markup cyclicity in response to MP shocks. The indirect effect determines the reallocation of costs across firms discussed in Section 4.2.4, but remains

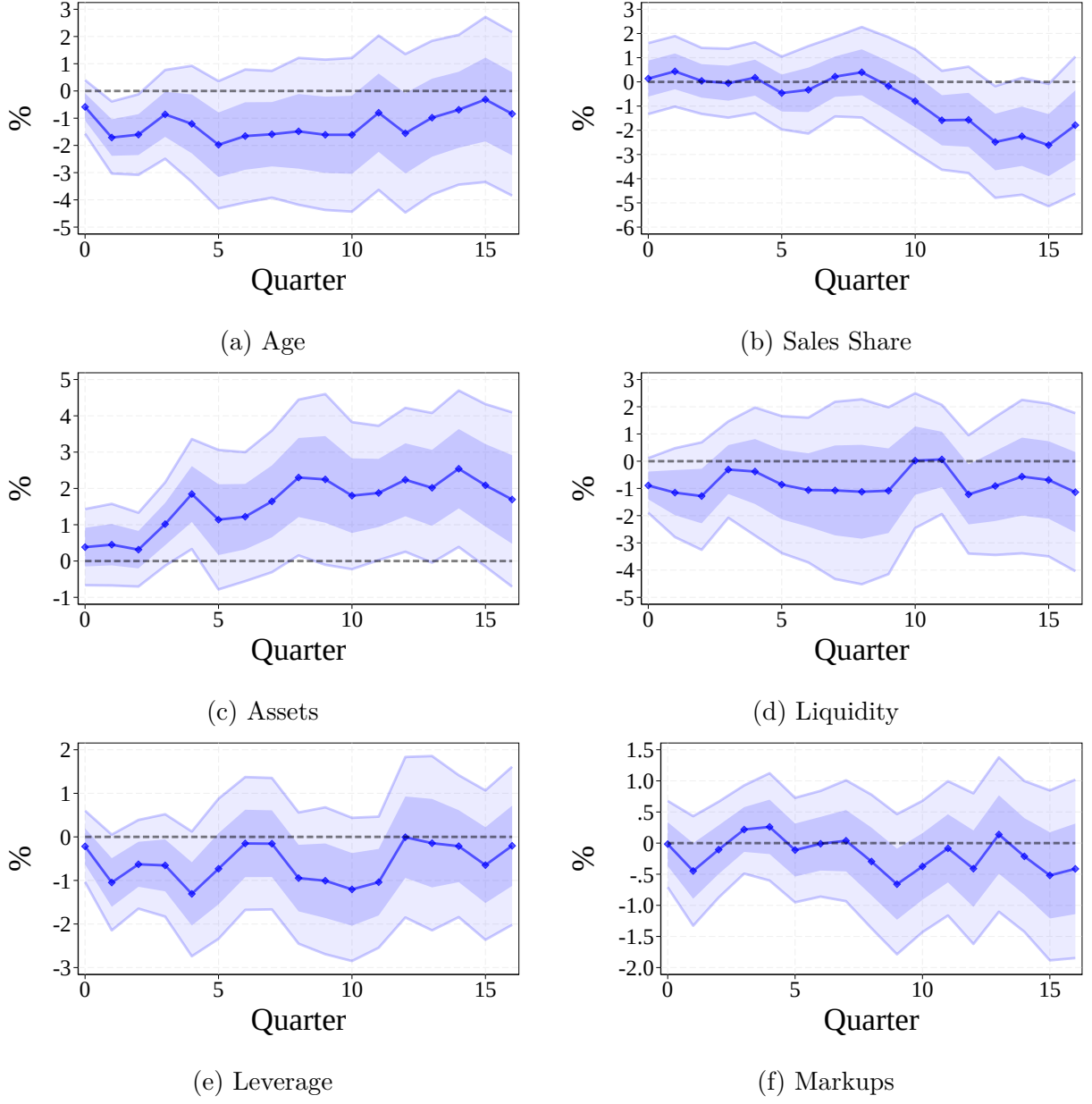
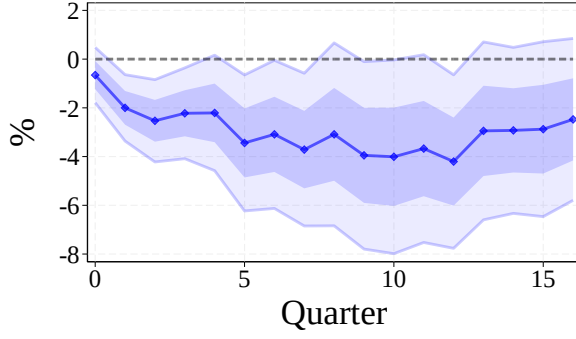
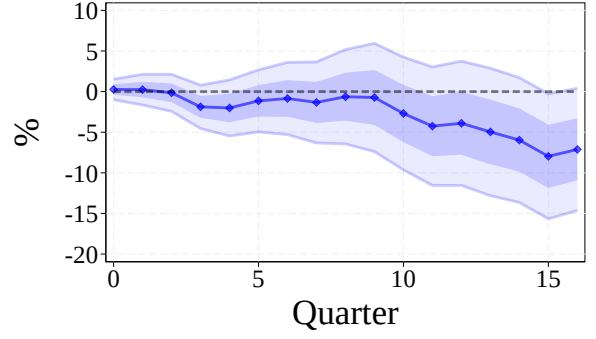


Figure A.17: Heterogeneous Sales Cyclicity By Firm-Level Characteristics
Note: Figure A.17 shows the relative sales response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Note that Figure A.17 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$ when estimating Equation (5) with all covariates. Standard errors clustered at the firm and quarter level.

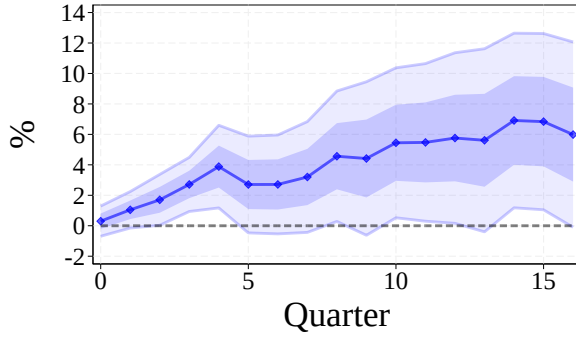
quantitatively weak. This is due to the fact that the markups of young and old firms, irrespective of the age measure used, are quite similar — i.e., $(\mu_O - \mu_Y)/\mathcal{M}$ is approximately



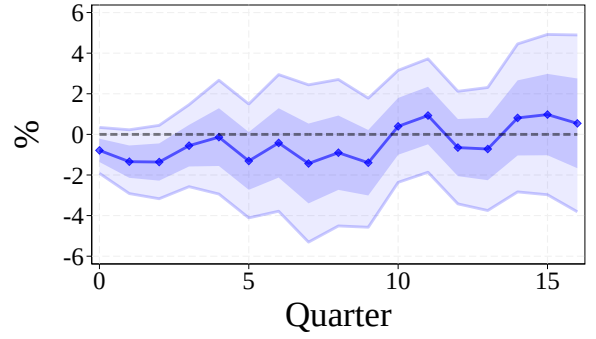
(a) Age



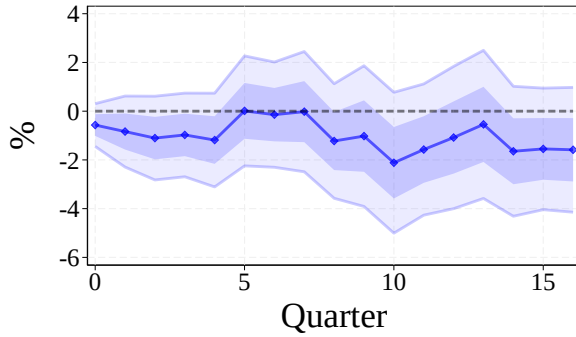
(b) Sales Share



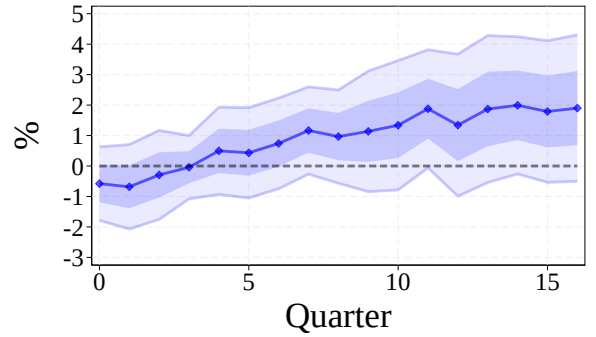
(c) Assets



(d) Liquidity



(e) Leverage



(f) Markups

Figure A.18: Heterogeneous Variable Costs Cyclicity By Firm-Level Characteristics
Note: Figure A.18 shows the relative variable costs response of (a) old vs young, (b) high vs low sales share, (c) big vs small, (d) high vs low liquidity, (e) high vs low leverage, (f) high vs low markup firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Note that Figure A.18 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$ when estimating Equation (5) with all covariates. Standard errors clustered at the firm and quarter level.

zero.

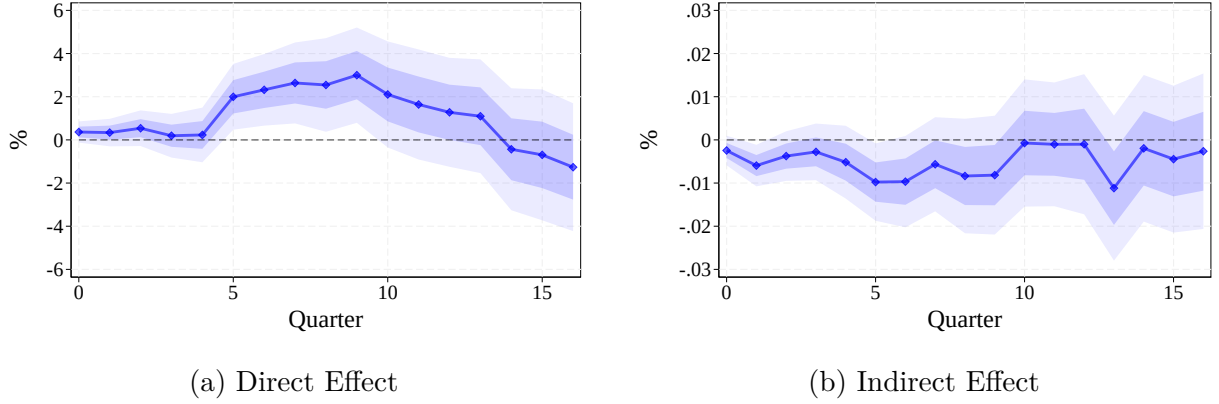


Figure A.19: Direct and Indirect Effects

Note: [Figure A.19a](#) and [Figure A.19b](#) present the direct and indirect effects as defined in Equation (9). The IRFs are normalized to a 25-basis-point contractionary MP shock. The dark blue line with squares represents the results obtained using our baseline measure of age. The dark and light blue shaded areas indicate the 68 percent and 90 percent confidence intervals around the point estimates, respectively.

A.3.2 Sales-Weighted Markup Aggregation

If we compute the aggregate markup as the sales — instead of the variable cost — weighted average of firm-level markups, we can examine how sales reallocate between young and old firms in response to MP shocks, as suggested by the conceptual framework outlined in Section 2. For this, we use Equation (5), but with firm-level sales shares as dependent variable on the left-hand side instead of firm-level (log) markups. Note that since the sum of sales responses to a MP shock between the two firm-age groups must be zero by construction, identifying the relative sales share response for old firms is equivalent to identifying the level of the sales-share response of old firms. [Figure A.20](#) illustrates the response of sales shares for older firms following a contractionary MP 25-basis-point shock and show a similar pattern to the one estimated for variable cost-shares in [Figure 3](#). Specifically, there is reallocation of sales (and variable costs) from old to young firms, mirroring the theoretical result in [Baqae, Farhi and Sangani \(2024\)](#), which describes a reallocation from large to small firms after a MP tightening.

Using our baseline measure of age, we find that the share of variable sales for firms above the median age, ω_O^s , is 0.76; the variable sales-weighted markups for older firms, μ_O , is 1.62;

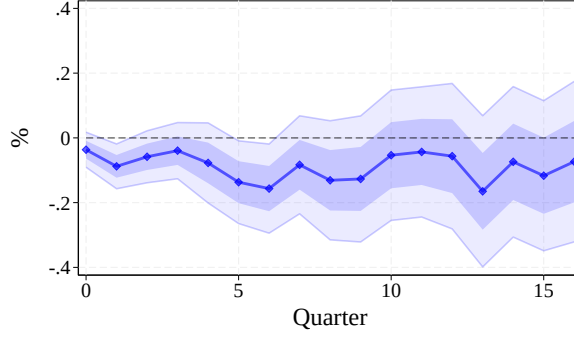


Figure A.20: Heterogeneous Sale Shares Cyclicalilty

Note: [Figure A.20](#) shows the variable sale shares response of old firms conditional on a MP shock. Coefficients $\widehat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Note that [Figure 3](#) is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

the variable sales-weighted markups for younger firms, μ_Y , is 1.57; and the aggregate markup using sales instead of costs-weights, $\mathcal{M}$, is 1.61. Notice that here the aggregate markups is higher than the one calculated with cost shares, as explained in [De Loecker et al. \(2020\)](#).

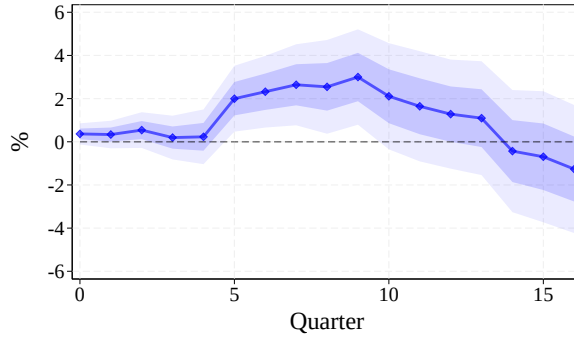


Figure A.21: Aggregate Sales-Weighted Markup Cyclicalilty

Note: [Figure A.21](#) presents the aggregate markup response conditional on a MP shock, derived using Equation (9). The results in [Figure A.21](#) are normalized to a 25-basis-point contractionary MP shock. The dark blue line with squares represents the results obtained using our baseline measure of age. The dark and light blue shaded areas indicate the 68 percent and 90 percent confidence intervals around the point estimates, respectively.

By combining these numbers with our IRF estimates from Section 4 and Equation (9), we now compute the IRF of the aggregate markup in response to a MP shock, as well as its decomposition by direct and indirect effect. [Figure A.21](#) shows that — conditional on a 25-

basis-point MP tightening — the estimated sales-weighted markup cyclicalities qualitatively and quantitatively reflect the cost-weighted markup cyclicalities presented in our baseline specification (Figure 4). Similarly, Figure A.22 confirms that the "direct effect", summarized by the heterogeneous markup responses across different firm-age groups following a MP shock, is what drives the conditional countercyclicalities of the aggregate markup. This is regardless of whether the aggregate markup is weighted using sales shares or cost shares, as in Figure A.19.

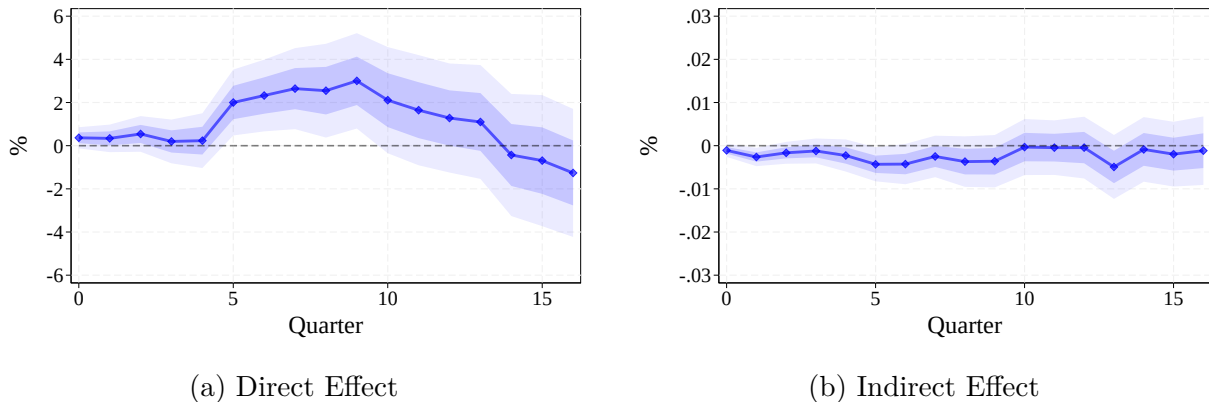


Figure A.22: Direct and Indirect Effects of Sales-Weighted Aggregate Markup
Note: Figure A.22a and Figure A.22b present the direct and indirect effects as defined in Equation (9). The IRFs are normalized to a 25-basis-point contractionary MP shock. The dark blue line with squares represents the results obtained using our baseline measure of age. The dark and light blue shaded areas indicate the 68 percent and 90 percent confidence intervals around the point estimates, respectively.

A.3.3 Changing Firm-Level Responses Over Time

In what follows, we estimate an alternative version of Equation (5) to show that firm-level markup cyclicalities across different age groups has not changed over time, as firms' differential responses before and after 2000 are estimated to be close to zero. Figure A.23 illustrates the results of this estimation and suggests therefore that the shift in the response of the aggregate markup to MP shocks over time should be driven by changes in the distribution of firms over time, and not by changes in the responsiveness of a particular group of firms over time.

$$\begin{aligned}
\Delta_h \log \mu_{i,t+h} = & \sum_{x \in \mathcal{X}} \left(\alpha_{x,h} + \beta_{x,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{x,h}^k \varepsilon_{t+k}^m \right) \times \mathbf{1}_{i \in \mathcal{I}^x} \\
& + \sum_{x \in \mathcal{X}} \left(\alpha_{t \geq 2000q4,x,h} + \beta_{t \geq 2000q4,x,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{t \geq 2000q4,x,h}^k \varepsilon_{t+k}^m \right) \\
& \times \mathbf{1}_{i \in \mathcal{I}^x} \times \mathbf{1}_{t \geq 2000q4} + \sum_{\ell=1}^L \delta'_h X_{i,t-\ell} + \varphi_{i,h} + \varphi_{s,t,h} + \vartheta_h t + u_{i,t+h}
\end{aligned}
\tag{A.4}$$

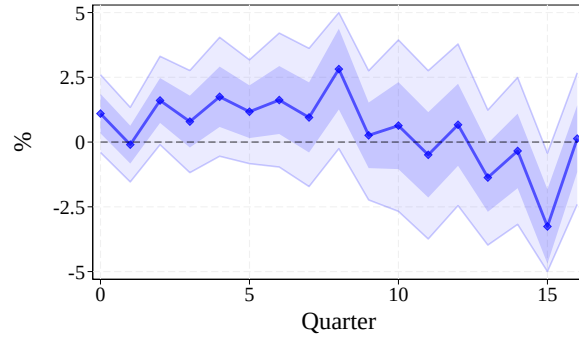


Figure A.23: Changing Firm-Level Responses Over Time

Note: Figure A.23 presents the estimate change before and after 2000 in the differential response of old relative to young firms. The results in Figure A.23 are normalized to a 25-basis-point contractionary MP shock. The dark blue line with squares represents the point estimates. The dark and light blue shaded areas indicate the 68 percent and 90 percent confidence intervals around the point estimates, respectively.

References

- ARELLANO, M. AND S. BOND, “Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations,” *The Review of Economic Studies* 58 (1991), 277–297.
- BAQAEE, D. R. AND E. FARHI, “Productivity and misallocation in general equilibrium,” *The Quarterly Journal of Economics* 135 (2020), 105–163.
- BAQAEE, D. R., E. FARHI AND K. SANGANI, “The supply-side effects of monetary policy,” *Journal of Political Economy* 132 (2024), 1065–1112.

- BLUNDELL, R. AND S. BOND, “Initial conditions and moment restrictions in dynamic panel data models,” *Journal of Econometrics* 87 (1998), 115–143.
- BOND, S., A. HASHEMI, G. KAPLAN AND P. ZOCH, “Some unpleasant markup arithmetic: Production function elasticities and their estimation from production data,” *Journal of Monetary Economics* 121 (2021), 1–14.
- DE LOECKER, J., J. EECKHOUT AND G. UNGER, “The rise of market power and the macroeconomic implications,” *The Quarterly Journal of Economics* 135 (2020), 561–644.
- FLYNN, Z., J. TRAINA AND A. GANDHI, “Measuring markups with production data,” *Available at SSRN 3358472* (2019).
- GANDHI, A., S. NAVARRO AND D. A. RIVERS, “On the identification of gross output production functions,” *Journal of Political Economy* 128 (2020), 2973–3016.
- GÜRKAYNAK, R. S., B. SACK AND E. SWANSON, “The sensitivity of long-term interest rates to economic news: Evidence and implications for macroeconomic models,” *American Economic Review* 95 (2005), 425–436.
- JAROCIŃSKI, M. AND P. KARADI, “Deconstructing monetary policy surprises—the role of information shocks,” *American Economic Journal: Macroeconomics* 12 (2020), 1–43.
- OTTONELLO, P. AND T. WINBERRY, “Financial heterogeneity and the investment channel of monetary policy,” *Econometrica* 88 (2020), 2473–2502.
- TEULINGS, C. N. AND N. ZUBANOV, “Is economic recovery a myth? Robust estimation of impulse responses,” *Journal of Applied Econometrics* 29 (2014), 497–514.